

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-061

WASTE DISCHARGE REQUIREMENTS
FOR
PINECREST PERMITEES ASSOCIATION, TUOLUMNE UTILITIES DISTRICT
AND UNITED STATES FOREST SERVICE
PINECREST WASTEWATER TREATMENT PLANT
TUOLUMNE COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. The Pinecrest Permittees Association (hereafter PPA) submitted a Report of Waste Discharge (RWD), dated 16 November 2000, for its wastewater treatment plant. PPA is responsible for the operation and management of the collection, treatment and disposal system that is owned by Tuolumne Utilities District (TUD). The collection, treatment and disposal system is situated on land owned by the United States Forest Service (USFS) and a portion of the sewage collection system is owned and administered by USFS. A recreational vehicle (RV) dump station, owned by USFS and situated on USFS property, discharges to the Pinecrest treatment plant. USFS has had a contract with TUD for treatment and disposal of the RV waste, however, the most recent contract has expired. Each entity shall hereafter be referred to individually as "PPA", "TUD", or "USFS", respectively, or jointly as "Discharger".
2. PPA, TUD, and USFS shall be responsible for compliance with these WDRs as they apply to the wastewater collection, treatment, and disposal system, while USFS shall be responsible for compliance in regards to the RV dump station.
3. The wastewater treatment and disposal facility is situated on USFS property at a latitude of 38° 10' 53" and a longitude of 120° 00' 23".
4. The wastewater treatment plant is near State Highway 108, approximately twenty-five miles northeast of Sonora in Section 28, T4N, R18E, MDM&M, as shown on Attachment A, which is attached hereto and made part of the Order by reference.
5. The RV dump station is just north of State Highway 108, approximately one half mile west of the Pinecrest turnoff, in Section 21, T4N, R18E, MDM&M.
6. Order No. 85-179, adopted by the Board on 28 June 1985, prescribes requirements for the PPA wastewater treatment plant. This Order is neither adequate nor consistent with the current plans and policies of the Board.
7. The wastewater collection system collects wastewater from the community of Pinecrest (approximately 350 residences), from private camps and resorts, and from USFS campgrounds, a ranger station, and the RV dump station. The plant influent flow rate is primarily driven by the population swings in the community. The permanent winter

population of the basin is less than 25 people. However, significant winter population increases occur on weekends. The peak season is from late June through late August.

8. The influent to the system is primarily domestic in nature. The current monthly average off-season flows to the treatment plant range from approximately 30,000 gallons per day (gpd) to approximately 60,000 gpd. The monthly average peak-season flows increase to approximately 150,000 gpd. The design flow of the treatment plant is 240,000 gpd.
9. The collection system consists of both gravity mains and force mains. The force mains service 29 pump stations. All but two of the lift stations are equipped with auto dialers as a spill prevention tool. The purpose of the auto-dialers is to alert PPA staff if a problem occurs and the liquid level in a lift station rises above a certain level. The remaining two lift stations are equipped with outside warning lights to alert passers-by of high liquid levels in the lift stations. The lift stations at the University of California's Camp Blue and Camp Gold are also equipped with 20,000-gallon overflow tanks, to permit the continued operation of the wastewater systems at the camps, even if problems occur at the lift stations.
10. The RV dump station is in service from May 1 to October 31 of each year. There is currently no means of measuring flow from the station. PPA has estimated that the average daily use throughout the six-month period is approximately 20 vehicles per day, and that peak use on busy Sundays is 50 to 100 vehicles per day. RV waste typically contains high strength waste. During peak times, wastewater from the RV dump station may cause slug loads that have the potential of upsetting the treatment plant.
11. The treatment plant provides secondary treatment with chlorination. The flow enters the plant's headworks, which consist of a Parshall flume and a mechanical bar rake. The flow then generally goes to the one million-gallon aerated ballast pond in the summer and to the ballast pond pump station in the winter (when the ballast pond is generally not in use). Flow is then directed to the plant proper, passing first through a roto-strainer with solids being deposited into the first of two aerobic digesters and the liquid entering the first chamber of a three-chamber rotating biological contactor (RBC). The RBC effluent then flows to a secondary clarifier. Solids from the secondary clarifier are transported to the first digester. Wastewater flows through a chlorine contact chamber where it is disinfected and is then pumped to one of the six evaporation/percolation ponds for disposal. Supernatant from either of the two digesters is returned to the headworks. Typically, the contents of the first digester are periodically transferred to the second digester for final treatment before being pumped to the sludge drying beds. The treatment plant layout is shown on Attachment B, which is attached hereto and made part of the Order by reference.
12. The ballast pond is the primary emergency wastewater storage facility. The pond is lined with a synthetic liner. During the summer months, the available storage capacity of the pond is typically 800,000 gallons or at least four days of summer flow. During the winter, the entire one million gallons is available for emergency purposes. A former USFS leachfield provides additional emergency disposal capacity of unknown volume. The intake to this field is located near the top of the ballast pond. Details regarding the leachfield system location and design are not available. Under normal usage, wastewater is pumped from the ballast pond to the treatment plant, but flow from the ballast pond can be directed to the evaporation/percolation ponds in emergency situations.

13. After the final treatment in the second digester, sludge is pumped to the sludge drying beds, which are located approximately 100 feet east of the treatment plant. The beds are located approximately 400 feet horizontally and 50 feet vertically from Camp Gold, a University of California summer camp. The beds cover approximately one-quarter acre and contain asphalt liners.
14. Debris and screenings from the headworks are placed in cans adjacent to the headworks to dewater, and are then transported to the Tuolumne County Transfer Station in Pinecrest. Sludge is bagged when dry and also transported to the Transfer Station for disposal.
15. From the treatment plant, treated effluent is pumped to one of six evaporation/percolation ponds for disposal. Ponds are identified as Ponds #1 through #6 as shown on Attachment B.
16. The evaporation/percolation ponds are located approximately 200 feet horizontally and 50 feet vertically from the North Fork of the Tuolumne River. The ponds are unlined and are constructed in soils described as the Ducey soil series, which are the product of glacial outwash deposition of transported granitic materials. The soil depth in the vicinity of the treatment plant is estimated to be 8 to 11 feet, at which depth soil grades into basement rock. The soil is generally coarse textured, with moderately rapid permeability, and described as a coarse sandy loam to sandy clay loam. Percolation rates in the vicinity reportedly averaged approximately 12 minutes per inch.
17. Because of concern over the potential for impact from the percolation ponds to the river, staff ordered monthly monitoring of the river in Fall 2000. Data collected to date is inconclusive. This Order requires continued monitoring and assessment of the river below the plant. If it is determined that surface water is not being impacted, groundwater monitoring may be required in the future to assess potential groundwater impacts below the site.
18. The evaporation/percolation pond system is operated by rotating effluent through the ponds to allow each of the pond bottoms to be mechanically ripped each year to optimize percolation. Pond spillways are sandbagged when necessary during winter months when total storage volume approaches the capacity of the ponds. Pond dikes are approximately one foot higher than the top of pond spillways.
19. During the fall of 2000, PPA deepened evaporation/percolation Ponds #3, #4, #5, and #6, thereby increasing their combined holding capacity by approximately 3.4 million gallons. The current holding capacity of the six evaporation/percolation ponds, with two feet of freeboard, is reportedly 10.13 million gallons. PPA is planning on deepening Ponds #1 and #2 after those ponds are drained during the summer of 2001. The anticipated holding capacity of the six ponds, after those modifications are complete, is unknown.
20. Based on PPA's RWD, the average biochemical oxygen demand (BOD) concentration of the wastewater influent to the plant is approximately 80 mg/l. Influent to the plant has not been analyzed for other constituents.
21. Based on PPA's RWD, the average effluent strength is reportedly:

Biochemical Oxygen Demand:	9 mg/l
Settleable Matter:	<0.1 mg/l
Total Coliform:	5 MPN/100ml
pH:	6.6 pH units

22. Because of heavy warm rains on heavy snow pack during the winter and/or spring of 1993, 1995, 1996, 1997, and 1998, eight spills of treated and disinfected effluent from the evaporation/percolation ponds to the North Fork of the Tuolumne River were reported. The volume spilled during four of the events was not reported. The volume of spills during the other four events ranged from two million to eight million gallons per event.
23. The University of California's Camp Gold is adjacent to and across the North Fork Tuolumne River from the plant. The river is heavily used by campers during summer months.
24. The staff at the University of California summer camps has reported that nuisance odors from the treatment plant often occur, generally increasing in severity as the camping season progresses. However, during the summer of 2000, complaints regarding nuisance odors were received as early as June. PPA took action to adjust the treatment process and mitigate the odors, and staff received no odor complaints after that time.
25. Federal regulations for stormwater discharges were promulgated by the U.S. Environmental Protection Agency on 16 November 1990 (40 CFR Parts 122, 123, and 124). The regulations require specific categories of facilities which discharge stormwater associated with industrial activities to obtain NPDES permits. The flow at this wastewater treatment plant is less than 1.0 mgd and therefore the Discharger is not required to apply for a stormwater NPDES permit.
26. Surrounding land uses are primarily recreational and open space.
27. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for waters of the Basins. These requirements implement the Basin Plan.
28. Surface water drainage is to the North Fork of the Tuolumne River.
29. The beneficial uses of the North Fork of the Tuolumne River are municipal and domestic supply, agricultural supply, power generation; contact and non-contact recreation; freshwater habitat; and wildlife habitat.
30. Specific information regarding groundwater quality in the vicinity of the treatment plant is not available. The beneficial uses of underlying groundwaters are municipal, industrial, and agricultural supply.
31. The Board has considered anti-degradation pursuant to State Board Resolution No. 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, this Order provides for data collection to determine whether the discharge will cause an increase in surface water constituents above that of background

levels. If the discharge is causing such an increase, then the Discharger may be required to cease the discharge, improve pond linings, implement source control, or take other action to prevent surface water degradation.

32. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
33. This discharge of wastewater is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a hazardous waste.
34. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."
35. The Board has notified the Discharger, and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
36. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 85-179 is rescinded and the Pinecrest Permittees Association, Tuolumne Utilities District, and United States Forest Service, their agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

(Note: Other prohibitions, conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991.)

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. The discharge shall not cause the degradation of any water supply.
5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
6. Surfacing of wastewater not collected and returned to the ballast pond is prohibited.
7. Discharge of untreated or partially treated waste to groundwater is prohibited.
8. Discharge to the USFS leachfield is prohibited.

B. Discharge Specifications:

1. The 30-day average dry weather flow into the treatment plant shall not exceed 170,000 gpd.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
3. As a means of discerning compliance with Discharge Specification No. 2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in all ponds and in the ballast pond shall not be less than 1.0 mg/l.
4. The ballast pond shall not have a pH of less than 6.0 or greater than 8.5.
5. The Dischargers shall operate all systems and equipment to maximize treatment of wastewater and optimize the quality of the discharge.
6. The ponds shall be managed to prevent the breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the waste surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, and/or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

7. The wastewater treatment system shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
8. The freeboard in evaporation/percolation ponds #2 through #6 shall never be less than one foot as measured vertically from the water surface to the lowest point on the top of the pond berm. Spillways may not be sandbagged to increase storage capacity. The freeboard in evaporation/ percolation pond #1 and the ballast pond shall never be less than two feet as measured vertically from the water surface to the lowest point on the top of the pond berm.

C. Effluent Limitations:

The discharge of effluent from the treatment plant in excess of the following limits is prohibited:

<u>Constituent</u>	<u>Units</u>	<u>30-Day Average</u>	<u>30-Day Median</u>	<u>Daily Maximum</u>
BOD	mg/l	30	--	60
Settleable Solids	ml/l	0.5		1.0
Total Coliform Organisms	MPN/100 ml	--	23	230

D. Solids Disposal Requirements:

1. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Storage, use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR Part 503.
3. Sludge and other solids shall be removed from ponds, digesters, etc. as needed to ensure optimal plant operation and adequate hydraulic capacity. Drying operations shall take place such that leachate does not impact groundwater or surface water.
4. If biosolids will be stored onsite between 15 October and 15 May of any year, then they shall be stored in a facility constructed in accordance with Class II surface impoundment or waste pile standards contained in Title 27 of the CCR, or as approved by the Executive Officer. Such a facility shall be designed and maintained to prevent inundation or washout from a storm or flood with a 100-year return frequency. PPA shall collect any leachate or stormwater that comes in contact with the biosolids pile and return it to the wastewater treatment plant.
5. Disposal of biosolids at a permitted municipal solid waste landfill or at a permitted publicly owned treatment works is acceptable. The Discharger may also elect to

dispose of its biosolids at a facility permitted under Order No. 2000-10-DWQ or at a similar facility permitted under individual WDRs. No matter where the biosolids are taken, the Discharger must comply with all sampling and analytical requirements of the entity that accepts the waste.

6. If the State Water Resources Control Board and the Regional Water Resources Control Board are given the authority to implement regulations contained in 40 CFR Part 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger shall comply with the standards and time schedules contained in 40 CFR Part 503 whether or not they have been incorporated into this Order.
7. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.

E. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentration statistically greater than background water quality, except for coliform bacteria. For coliform bacteria, increases shall not cause the most probable number of total coliform organisms to exceed 2.2 MPN/100 ml over any 7-day period.

F. Surface Water Limitations:

The discharge shall not cause the North Fork of the Tuolumne River downstream of the wastewater treatment plant to contain waste constituents in concentrations statistically greater than background (i.e., upstream) surface water quality.

G. Provisions:

1. All of the following reports shall be submitted pursuant to Section 13267 of the California Water Code:
 - a. **By 1 June 2001**, USFS shall submit an RV waste flow equalization plan to eliminate slug loads from the RV dump station during periods of peak use, along with a timeline for implementation of the plan. If the plan includes modifications that will require operations and maintenance expertise to ensure their proper functioning, an Operation and Maintenance Manual shall be prepared for DRSR or the subsequent operator. Interim measures to control slug loading that are feasible shall be implemented by **1 June 2001**. Full implementation of the plan shall be achieved by **15 November 2002**.
 - b. **By 1 June 2001**, PPA and TUD shall submit an Odor Mitigation Plan to prevent the occurrence of nuisance odor conditions. The plan shall include consideration of modification of the treatment system and moving the location of the sludge drying beds, and shall include a timeline for implementation of the plan.

- c. By **1 July 2001**, PPA and TUD shall submit a report which analyzes the inadequacies in the wastewater storage and disposal system that are causing the capacity limitation resulting in the spills described in Finding No. 22, and which limit the Discharger's ability to comply with Discharge Prohibition No. 1 and 2 and Discharge Specifications No. 8 at current flows. The report shall be prepared by a registered engineer, shall include a water balance, and shall evaluate the capacity of the ballast pond and evaporation/percolation ponds versus permitted flow limits and projected weather conditions.

The water balance shall calculate the monthly net flow volume into the system. Net flow shall include permitted inflow, I/I, seepage, design seasonal precipitation based on total annual precipitation using a 100-year return period (distributed monthly in accordance with historical precipitation patterns), normal evaporation rates, and measured percolation rates. The water balance shall take into account the fact that, in general, during the winter, percolation from the ponds diminishes significantly, and much of the flow into the treatment plant and precipitation is stored in the ponds until the spring thaws occur.

The report shall also propose necessary modifications or additions to the system to correct any inadequacies, and shall provide a proposed timeline for the improvements. The timeline shall include all actions necessary to comply with CEQA and submittal of an RWD, if necessary.

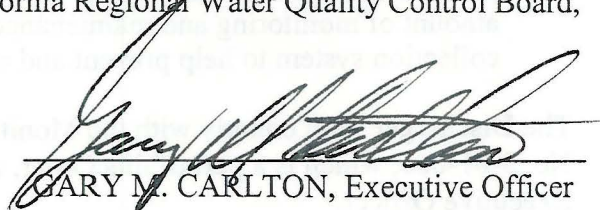
If, based on this report, system improvements are found to be necessary in order to comply with Discharge Prohibitions No. 1 and 2 and Discharge Specification No. 8, and those improvements are not completed by **15 October 2001**, then by that date, PPA and TUD shall provide a contingency plan describing interim measures that will be taken in order to assure compliance until long-term improvements can be made.

- d. By **1 August 2001**, PPA and TUD shall submit a collection system spill prevention plan, which describes the backup and/or emergency notification equipment in place or proposed at all lift stations in addition to warning lights to prevent future discharges of waste. The emergency notification equipment shall be capable of notifying personnel during evenings, weekends and holidays of potential problems at the lift stations. In addition, the report must define the amount of monitoring and maintenance to be performed annually on the collection system to help prevent and control accidental discharges.
2. The Discharger shall comply with the Monitoring and Reporting Program No. 5-01-061, which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.
3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the wastewater collection, treatment, or disposal system, the Discharger shall

notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

5. The Discharger shall submit to the Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.
6. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
7. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
8. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by TUD and/or the USFS, then TUD and/or the USFS shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
9. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
11. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 16 March 2001.


GARY M. CARLTON, Executive Officer

Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-01-061
FOR

PINECREST PERMITTEES ASSOCIATION, TUOLUMNE UTILITIES DISTRICT
AND UNITED STATES FOREST SERVICE
PINECREST WASTEWATER TREATMENT PLANT
TUOLUMNE COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring influent, effluent, ponds, surface water, and the RV dump station. This MRP is issued pursuant to Water Code Section 13267.

Pinecrest Permittees Association (PPA) and Tuolumne Utilities District (TUD) shall be responsible for implementation of this MRP as it pertains to the wastewater treatment plant and ponds. The United States Forest Service (USFS) shall be responsible for implementation of this MRP as it pertains to the RV dump station. Dischargers shall not implement any changes to the MRP unless and until a revised MRP is issued by the Executive Officer. Specific sample station locations shall be approved by Regional Board staff prior to implementation of sampling activities.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form.

INFLUENT MONITORING

Samples shall be collected at the same frequency and at approximately the same time as effluent samples and should be representative of the influent for the sampling period. Influent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	Gpd	Continuous	Daily	Monthly
BOD ₅ ¹	mg/l	Grab	Monthly	Monthly
Total Suspended Solids	mg/l	Grab	Monthly	Monthly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

EFFLUENT MONITORING

Wastewater effluent samples should be representative of the volume and nature of the discharge and shall be collected from the last connection through which waste can be admitted prior to discharge to the evaporation/percolation ponds. Grab samples are considered adequately composited to represent the effluent. Effluent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	Gpd	Continuous	Daily	Monthly
BOD ₅ ¹	mg/l	Grab	Weekly	Monthly
Total Settleable Solids	ml/l•hr	Grab	Weekly	Monthly
Total Coliform Organisms ²	MPN/100 ml	Grab	Weekly	Monthly
Total Suspended Solids	mg/l	Grab	Weekly	Monthly
PH	pH units	Grab	Weekly	Monthly
Total Dissolved Solids	mg/l	Grab	Monthly	Monthly
Nitrates as Nitrogen	mg/l	Grab	Monthly	Monthly
Formaldehyde	mg/l	Grab	Monthly	Monthly
Phenols	mg/l	Grab	Monthly	Monthly
Zinc	mg/l	Grab	Monthly	Monthly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

² Using a minimum of 15 tubes or three dilutions.

POND MONITORING

Samples shall be collected from established sampling stations located in areas that will provide a sample representative of the wastewater in the evaporation/equalization ponds and the ballast pond. Freeboard will be measured vertically from the surface of the pond water to the spillway, and shall be measured to the nearest 0.25 feet. Pond monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Freeboard	Feet	Measurement	Daily	Monthly
Dissolved Oxygen	mg/l	Grab	Weekly	Monthly

SURFACE WATER MONITORING

PPA and TUD shall establish two sampling stations in the North Fork Tuolumne River: one station shall be located approximately 50 feet upstream of the treatment plant boundary and one station approximately 50 feet downstream of the treatment plant boundary. An additional sampling station shall be established in the unnamed stream that enters the North Fork Tuolumne River just downstream of the Camp Gold dining room. Surface water samples shall be analyzed for the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Total Dissolved Solids	mg/l	Grab	Monthly
Chlorine Residual	mg/l	Grab	Monthly
Ammonia as Nitrogen	mg/l	Grab	Monthly
Nitrates as Nitrogen	mg/l	Grab	Monthly
Total Coliform Organisms ¹	MPN/100ml	Grab	Monthly

¹ Using a minimum of 15 tubes or three dilutions.

BIOSOLIDS MONITORING

PPA and TUD shall keep records regarding the quantity of biosolids generated by the treatment processes; any sampling and analytical data; the quantity of biosolids stored on site; and the quantity removed for disposal. The records shall also indicate that steps taken to reduce odor and other nuisance conditions. Records shall be stored onsite and available for review during inspections.

If biosolids are transported off-site for disposal, then the Discharger shall submit records identifying the hauling company, the amount of biosolids transported, the date removed from the facility, the location of disposal, and copies of all analytical data required by the entity accepting the waste. If biosolids are disposed of onsite, then the Discharger shall submit the annual report information as contained in the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).

All records shall be submitted as part of the Annual Monitoring Report.

RV DUMP STATION MONITORING

The United States Forest Service (USFS) shall monitor flow rates from the RV dump station to the treatment plant on a daily basis. USFS shall submit the data to the Board on a monthly basis.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, surface water, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

A. Monthly Monitoring Reports

PPA and TUD shall submit monthly reports to the Regional Board on the **1st day of the second month following sampling** (i.e. the January report is due by 1 March). At a minimum, the reports shall include:

1. Results of influent, effluent, pond, and surface water monitoring. Data shall be presented in tabular format.
2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements.
3. If requested by staff, copies of laboratory analytical report(s).
4. A calibration log verifying weekly calibration of field monitoring instruments (DO, pH, etc. meters) used to collect reported data.

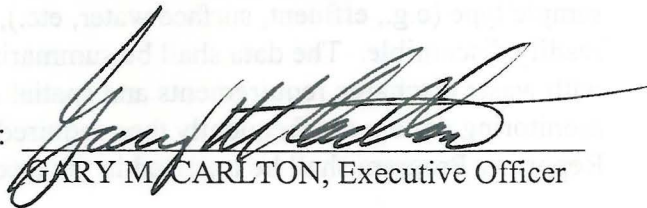
B. Annual Monitoring Report

PPA and TUD shall prepare an Annual Report as part of the December monitoring report. The Annual Report will include all monitoring data required in the monthly schedule. The Annual Report shall be submitted to the Regional Board by **1 February** of each year and shall include the following:

1. If requested by staff, tabular and graphical summaries of all data collected during the year.
2. The anticipated schedule for cleaning, drying, and disposal of biosolids;
3. Summary of information on the disposal of biosolids as described in the "Biosolids Monitoring" section, above.
4. An evaluation of the performance of the wastewater treatment and disposal systems, as well as a forecast of the flows anticipated in the next year.
5. An evaluation of the surface water adjacent to the treatment plant site.
6. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.
7. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

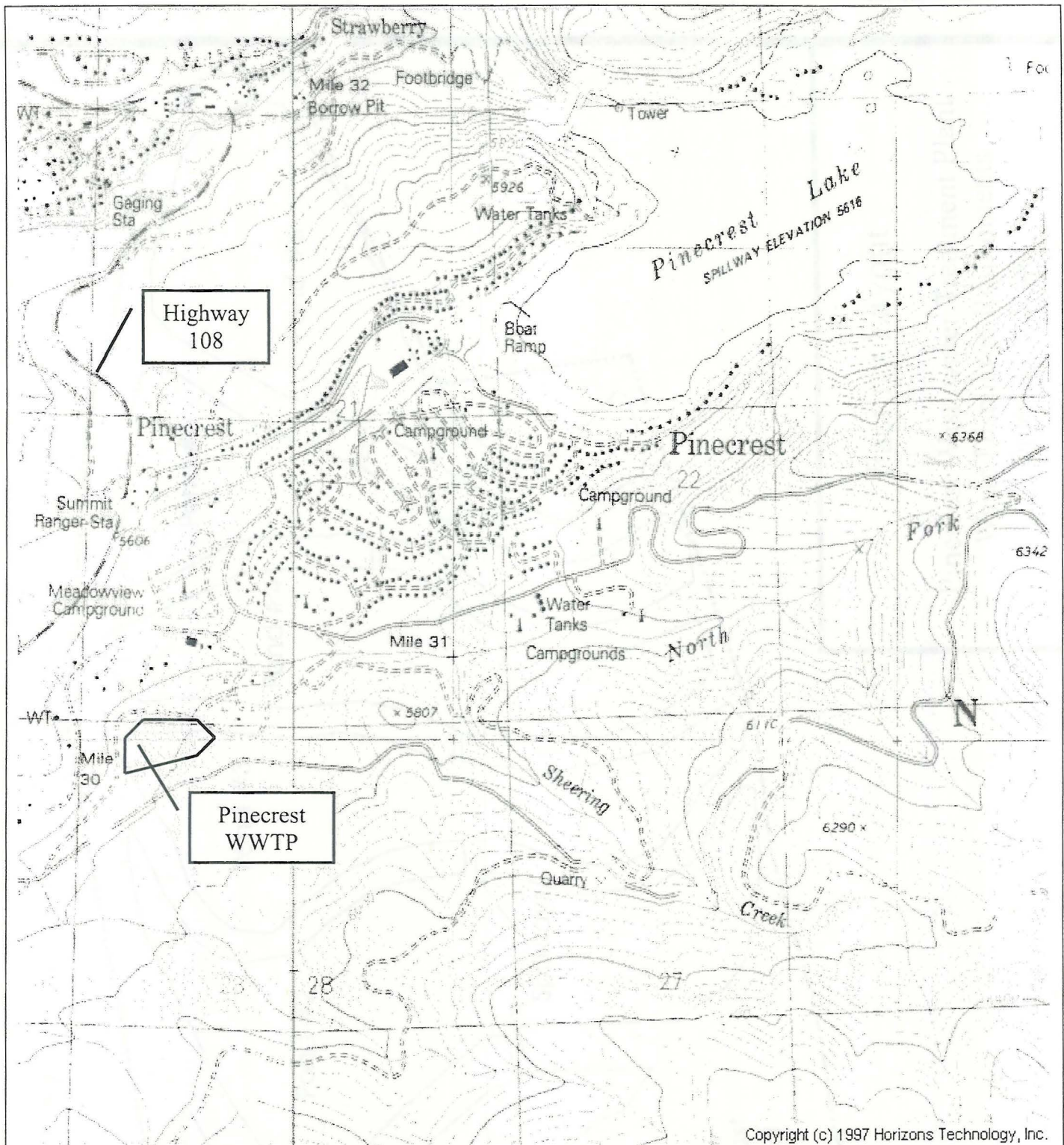
The Dischargers shall implement the monitoring program as of the date of this Order.

Ordered by:


GARY M. CARLTON, Executive Officer

16 March 2001

(Date)

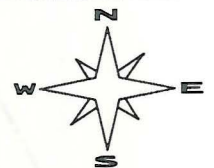


Drawing Reference:

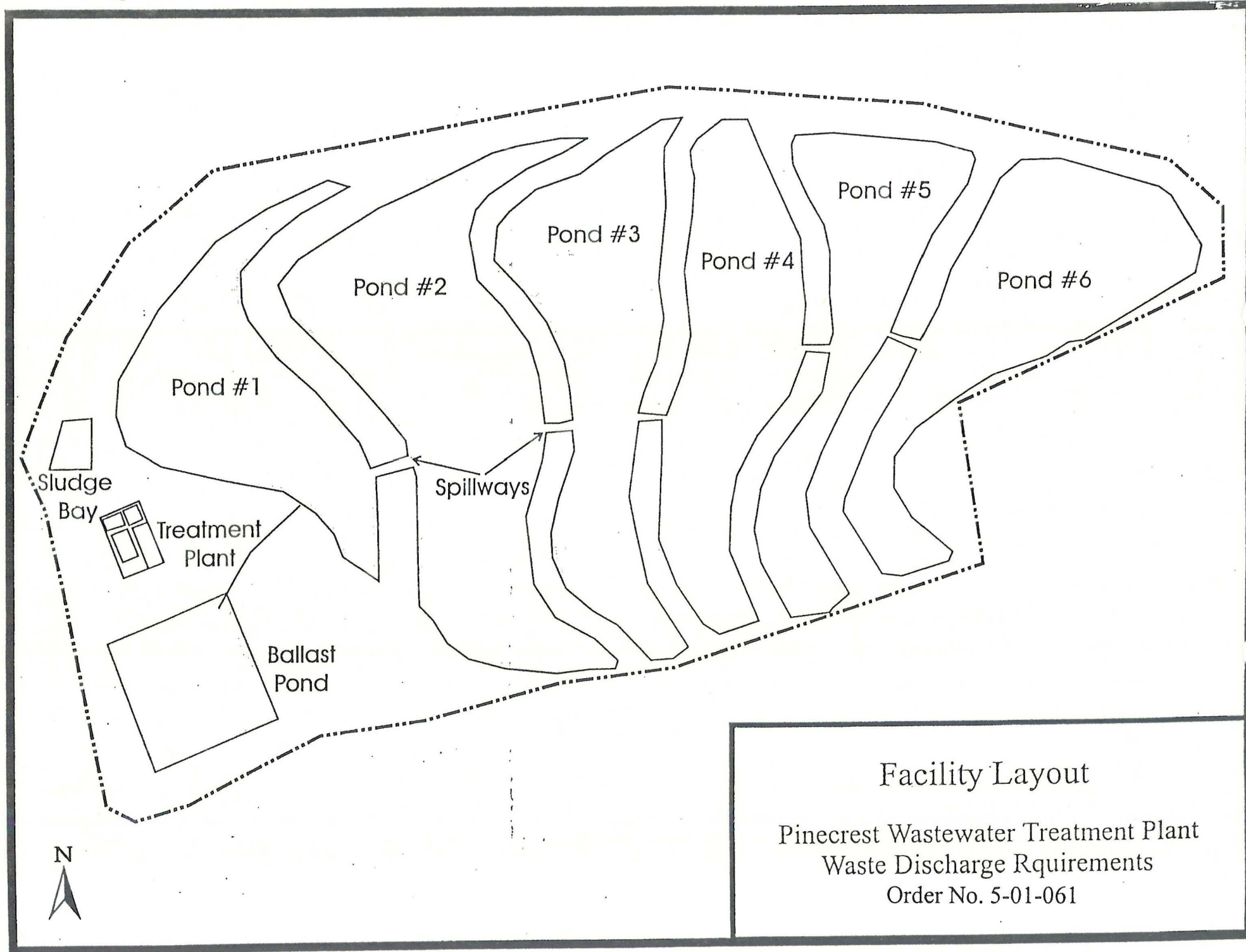
PINECREST QUADRANGLE
U.S.G.S TOPOGRAPHIC MAP
7.5 MINUTE QUADRANGLE

SITE LOCATION MAP

PINECREST WASTEWATER TREATMENT PLANT
WASTE DISCHARGE REQUIREMENTS
ORDER NO. 5-01-061



approx. scale
1 in. = 1,800 ft.



INFORMATION SHEET

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-01-061
PINECREST PERMITTEES ASSOCIATION, TUOLUMNE UTILITIES
DISTRICT, AND UNITED STATES FOREST SERVICE
PINECREST WASTEWATER TREATMENT PLANT
TUOLUMNE COUNTY

The Pinecrest Permittees Association (PPA) operates a wastewater treatment plant that serves the needs of the community of Pinecrest. PPA is a separate and distinct entity that operates and manages the wastewater collection, treatment, and disposal system under contract with the system owner, Tuolumne Utilities District (TUD). The treatment and disposal facility is situated on land owned by the United States Forest Service (USFS). A recreational vehicle (RV) dump station, owned by the USFS, discharges to the Pinecrest treatment plant. PPA shall be primarily responsible for compliance with these WDRs as they apply to the wastewater collection, treatment, and disposal system, while USFS shall be responsible for compliance in regards to the RV dump station. The wastewater treatment plant is located near State Highway 108, approximately twenty-five miles northeast of Sonora. The treatment plant provides secondary treatment with chlorination. The current monthly average off-season flows to the treatment plant range from approximately 30,000 gallons per day (gpd) to approximately 60,000 gpd. The monthly average peak-season flows increase to approximately 150,000 gpd. The design flow of the treatment plant is 240,000 gpd.

Treated effluent is pumped to one of six evaporation/percolation ponds for disposal. The evaporation/percolation ponds are located approximately 200 feet horizontally and 50 feet vertically from the North Fork of the Tuolumne River. The ponds are unlined and are constructed in soils described as a coarse sandy loam, which are highly permeable. Because of heavy warm rains on heavy snow pack during the winter and/or spring, eight spills of treated and disinfected effluent from the evaporation/percolation ponds to the North Fork of the Tuolumne River were reported in five of the past eight years. In order to address this problem, the WDRs contain a time schedule to analyze the problem, propose modifications to correct the problem, and implement the modifications.

The staff at the University of California summer camps, situated adjacent to the plant, has reported that nuisance odors from the treatment plant often occur, generally increasing in severity as the camping season progresses. In order to address this problem, the WDRs contain a time schedule to submit an Odor Mitigation Plan to prevent the occurrence of nuisance odor conditions in the future.

The USFS RV dump station is in service from May 1 to October 31 of each year. Although flows from the station to the treatment plant are not measured, it is believed that peak flows of two to five times normal occur on Sundays. Because RV waste typically contains very high strength waste, these peak flows may cause slug loads that have the potential of upsetting the treatment plant. In order to address this problem, the WDRs contain a time schedule for the USFS to submit a plan to equalize flows from the dump station and to implement the plan.

The Monitoring and Reporting Program includes requirements for monitoring of influent and effluent, the ballast pond, surface water, biosolids, and the RV dump station.